



CSCE 633-700: Machine Learning

Linear regression

Overview

Linear Regression

- Example
- Representation, Evaluation
- Optimization: Closed form solution via Ordinary Least Squares
- Optimization: Numerical solution via Gradient Descent
 - General gradient descent
 - Gradient descent for linear regression (batch, stochastic, mini-batch)
- Non-linear basis function for regression & Overfitting

[Parts of these slides have been adapted from K. Murphy (Machine Learning: A probabilistic perspective), Dr. Andrew Ng's Machine Learning course at Coursera, and CSCI567 Machine Learning (USC, Drs. Sha & Liu)]

Linear Regression: Representation

- **Input:** $\mathbf{x} \in \mathbb{R}^{D+1}$ (D covariates/predictors/features, 1 extra term in the first position that corresponds to the bias term)
- **Output:** $y \in \mathbb{R}$ (responses, targets, outcomes, etc.)
- **Training Data:** $\mathcal{D} = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$
- **Model:** $f: \mathbf{x} \rightarrow y, f(\mathbf{x}) = w_0 + \sum_{d=1}^D w_d x_d = \mathbf{w}^T \mathbf{x}$
 w_0 : bias term
 $\mathbf{w} = [w_0, w_1, \dots, w_D]^T \in \mathbb{R}^{D+1}$: parameters/weights $\mathbf{x}_n = [1, x_{n1}, \dots, x_{nD}]^T \in \mathbb{R}^{D+1}$: n^{th} training sample